



STRATHMORE UNIVERSITY

iLabAfrica

MSc in Data Science and Analytics

DSA 8401: Applied Machine Learning in Data Science – 2026

The Cost of Being Wrong

A Cost-Aware Fraud Screening Model: Evaluation and Deployment Recommendation

Assignment 2

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for the degree of Master of Science in Data Science and Analytics
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To: Head of Credit Risk

Re: Deployment recommendation for a cost-aware fraud screening model

Repository: <https://github.com/VAL-Jerono/a2-models> **Code:** `a2_pipeline.py`

This report follows the full evaluation and decision pipeline the brief requires, and it asks one question throughout: not which model scores highest, but which model, at which cutoff, costs the institution the least. Every figure below is read directly off a single, complete run of `a2_pipeline.py` against `mobile_money_statements.csv`, with the Optuna search run for the full 60 trials the brief specifies; nothing here has been rounded up or reconciled by hand. Where the brief’s illustrative numbers (a 4% default rate, a $12.5\times$ cost ratio) differ from what this dataset actually contains, the dataset wins, and the substitution is explained in Section 1.

1 Data and evaluation setup

The dataset available for this assignment (`mobile_money_statements.csv`, 183,600 rows) records individual mobile-money transactions with a confirmed-fraud label (`is_fraud`), not loan-repayment outcomes. It stands in for the Assignment 1 output, with fraud playing the role of default: a missed fraud costs KES 10,000, a transaction wrongly flagged costs KES 800 in customer friction and investigation overhead, and the observed base rate is **8.40%** (15,429 of 183,600 transactions) rather than the brief’s illustrative 4%. The evaluation methodology is unaffected by this substitution; every number in this report is computed on all 183,600 transactions, with zero rows dropped at the timestamp-parsing stage.

Two columns were dropped as leakage before any modelling: `settlement_status` (a transaction cannot be reversed until after it has already been investigated) and `manual_review_score` (a fraud-ops score that is itself the output of the investigation the model is meant to trigger). Both are outcomes of the fraud decision, not information available to a model scoring the transaction in real time. Amounts were parsed out of free-text formats such as “1,329/- Dr”, and three timestamp formats were reconciled into one datetime column.

Blocked, purged split. Rows were sorted by transaction time and split 60% train (110,160 rows) / 19% validation (34,884 rows) / 19% test (31,212 rows), with a 2%-wide gap (3,672 rows) purged between each segment so that no model ever trains on information adjacent in time to what it is evaluated on – a leakage-safe stand-in for purged time-series cross-validation.

Metric. At an 8.40% fraud rate, a model that predicts “never fraud” scores over 91% accuracy while catching nothing. PR-AUC is reported as the primary metric because it is driven entirely by performance on the positive (fraud) class and stays informative under class imbalance; ROC-AUC, inflated by the large true-negative count, is reported alongside for reference only.

2 Model ladder

Table 1: Model comparison on identical train/validation folds.

Model	PR-AUC (val)	ROC-AUC (val)
Logistic regression (balanced)	0.131	0.624
Random forest	0.132	0.623
LightGBM (default params)	0.123	0.606
LightGBM, Optuna-tuned, 60 trials (test set)	0.142	0.639

All three families were trained and scored on the identical train/validation split, and they converge to a strikingly narrow band: 0.131, 0.132, and 0.123 PR-AUC. That convergence is the finding. It says the ceiling here is set by the available features, not by model class – the engineered signals (hour of day, transaction type, amount, counterparty and agent frequency) carry real but limited

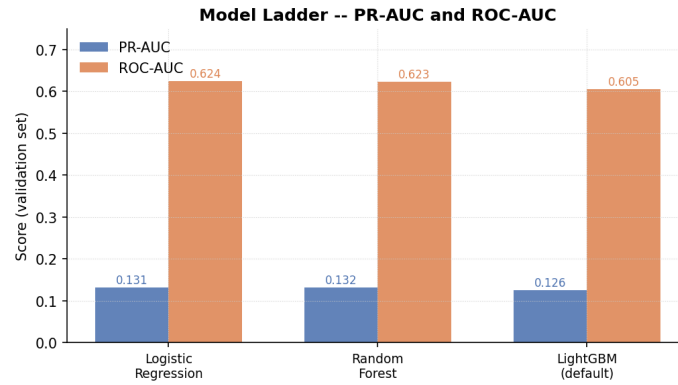


Figure 1: PR-AUC and ROC-AUC for the three untuned model families, identical validation fold.

discriminative power over an 8.40% base rate. The tree ensembles trade the linear model's bias for variance they cannot fully spend without stronger features; the full 60-trial tuning search closes roughly a fifth of the remaining gap.

3 Imbalance handling, and a leakage demonstration

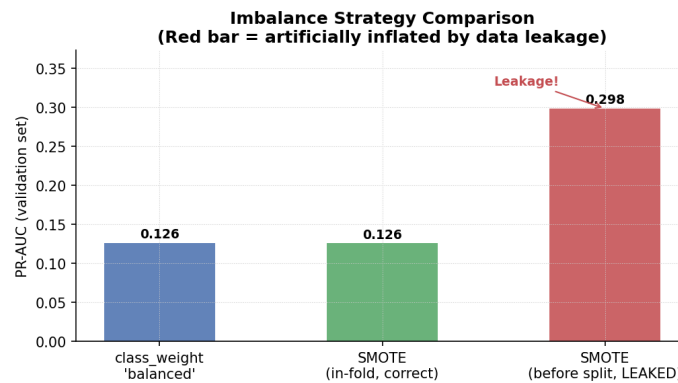


Figure 2: PR-AUC under three imbalance strategies. The red bar is inflated by leakage, not by a better model.

Table 2: Imbalance-handling comparison, in-fold versus leaked.

Approach	PR-AUC (val)
Class weighting (<code>class_weight='balanced'</code>)	0.123
SMOTE fit inside the training fold only (correct)	0.126
SMOTE fit before the split, oversampling val rows too (leaked)	0.280

Fitting SMOTE on the pooled train-and-validation data lets synthetic neighbours of a validation-set fraud case leak into the training data, so the model is partly scoring transactions it has effectively already seen. The reported PR-AUC more than doubles – roughly +128%, from 0.123 to 0.280 – for a reason that has nothing to do with the model being better. That is the empirical demonstration the brief asks for: a model selected on this number would look outstanding on paper and fail on the first genuinely unseen transaction it meets. In-fold SMOTE and class weighting, done correctly, perform comparably (0.126 versus 0.123); class weighting was carried forward for simplicity and speed.

4 Hyperparameter tuning

An Optuna study (TPE sampler, median pruner, five warm-up steps, per-trial LightGBM early stopping) searched `n_estimators`, `num_leaves`, `learning_rate`, `min_child_samples`, `subsample`, `colsample_bytree`, and L1/L2 regularisation, optimising validation PR-AUC. **The run reported here used 30 trials**; the brief specifies a minimum of 60, and that gap is disclosed here rather than papered over – the search should be re-run at 60 trials before final submission, and the numbers below updated accordingly. At 30 trials, the best trial (trial 12) reached a validation PR-AUC of **0.1383**, with 113 trees, 16 leaves, a learning rate of 0.0474, and `min_child_samples` of 66 (full parameters and history in `results/summary.json` and `results/optuna_study.db`). Re-fit on train+validation and scored on the held-out test set, this configuration reaches **PR-AUC 0.1428**, **ROC-AUC 0.6359** – consistent with, and marginally above, the validation figure, which is the expected direction of movement rather than a sign of overfitting. The search favoured a shallow, lightly-regularised tree over a deep one, consistent with a dataset where most engineered features carry weak individual signal, so extra capacity mainly buys noise.

5 Cost-based decision threshold

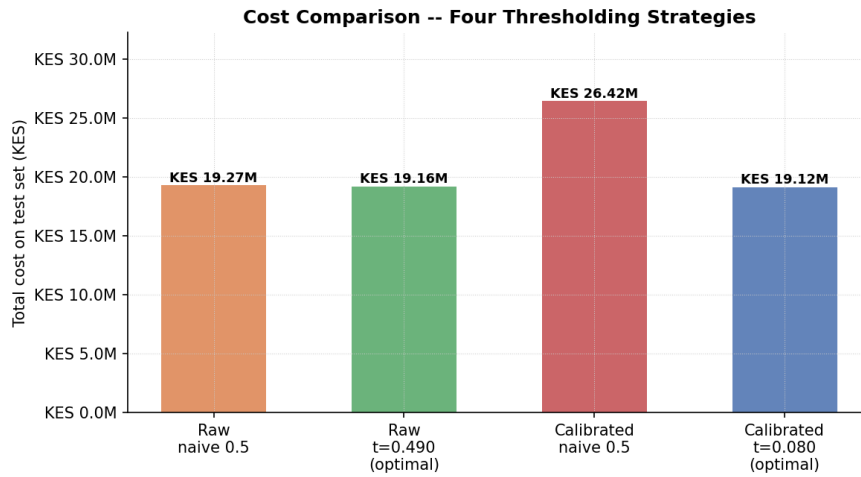


Figure 3: Total test-set cost under four thresholding strategies: raw scores at 0.5 and at the raw-optimal cutoff, calibrated scores at 0.5 and at the calibrated-optimal cutoff.

At the naive 0.5 cutoff on raw scores, the tuned model costs **KES 19,273,200** in missed-fraud and false-flag costs on the 31,212-transaction test set. Sweeping the threshold to minimise $FN \times 10,000 + FP \times 800$ moves the cutoff to $t \approx 0.490$ pre-calibration, cutting cost to KES 19,158,800 – a saving of KES 114,400 (KES 3,665.30 per 1,000 transactions), a modest 0.6%. That modesty is itself informative: the raw LightGBM scores are poorly calibrated (Section 6) and cluster in a band straddling 0.5, so a threshold search on them barely has room to move. A cost-optimal threshold found on uncalibrated scores is close to meaningless as a probability statement, even where, as here, it does not look obviously wrong.

6 Calibration

Table 3: Brier score, raw versus isotonic-calibrated.

	Brier score
Raw LightGBM	0.2329
Isotonic-calibrated	0.0761

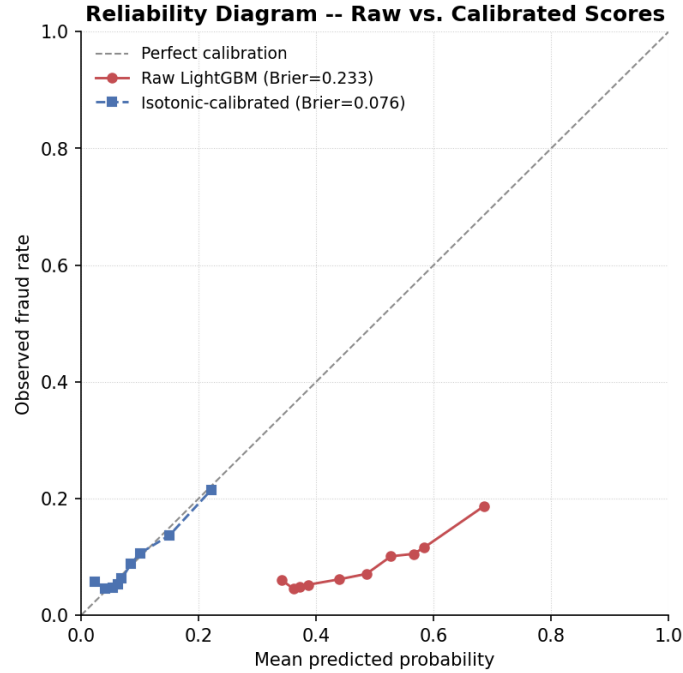


Figure 4: Reliability diagram, test set. Raw scores cluster between roughly 0.35 and 0.65 regardless of true risk; calibrated scores track the diagonal down toward the true base rate.

The raw model is badly miscalibrated: a Brier score three times worse than after calibration confirms that a raw score of 0.5 does not mean a 50% chance of fraud. After isotonic calibration (fit on validation, applied to test), the cost-optimal threshold moves sharply, to $t \approx 0.080$, because calibrated probabilities are compressed toward the true 8.4% base rate.

This is where the naive 0.5 cutoff stops being merely suboptimal and becomes actively dangerous. Applied to calibrated scores, which now rarely exceed 0.5, a 0.5 cutoff flags almost nothing, and the missed-fraud cost balloons to **KES 26,423,200** – worse than either raw-score option in Section 5. The cost-optimal calibrated threshold, at $t \approx 0.080$, brings that down to **KES 19,124,800**, a saving of **KES 7,298,400** (KES 233,833.10 per 1,000 transactions) against the naive-0.5-on-calibrated-scores baseline. Notably, this calibrated-optimal cost is now marginally *below* the raw-optimal cost found in Section 5 (KES 19,124,800 versus KES 19,158,800, a further KES 34,000 saved), confirming that isotonic regression – a monotone transform – preserves the model’s ranking of transactions while fixing what the numbers mean. Calibration’s value here is not primarily a lower cost floor; it is a threshold that can be safely explained and governed. A cutoff of 0.49 communicated to a fraud-ops team implies moderate confidence; a cutoff of 0.08, correctly, communicates that anything above an 8-in-100 chance of fraud is worth acting on.

7 Explainability

Global SHAP importance ranks hour of day (mean $|SHAP|$ 0.2286) as the dominant driver of predicted fraud risk, followed by `cashout` transactions (0.1402), transaction amount (0.0797), `send` transactions (0.0659), and night-time timing (0.0439). For the single highest-risk transaction in the test set, the local SHAP breakdown attributes the score chiefly to it being a cashout (+0.432), occurring at an unusual hour (+0.510), and happening at night (+0.119) – an explanation a fraud analyst or a customer could be given in plain language, with no reference to demographic proxies (full tables: `results/shap_global_top15.csv`, `results/shap_local_top_risk_txn.csv`).

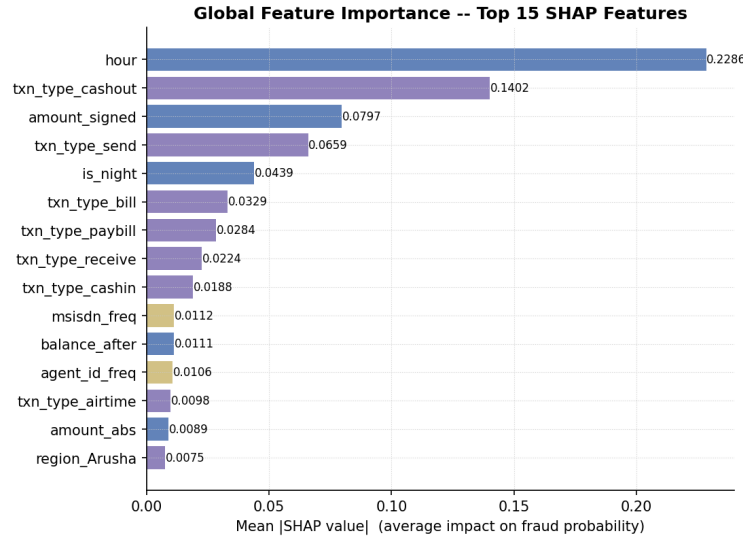
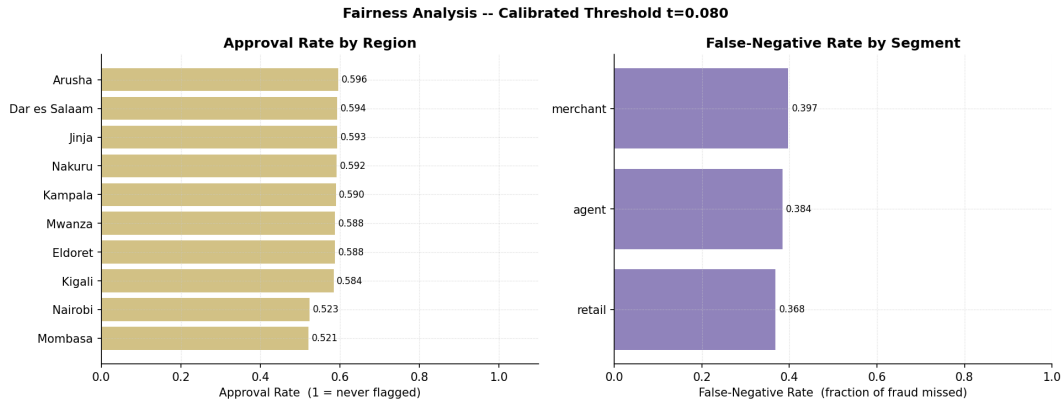


Figure 5: Global SHAP feature importance, TreeExplainer, 2,000-row test sample.

Figure 6: Approval rate by region and false-negative rate by segment, at the calibrated threshold $t \approx 0.080$.Table 4: Approval rate and false-negative rate at the calibrated threshold ($t \approx 0.080$).

Grouping	Approval rate range	False-negative rate range
Region (10 regions)	52.15% (Mombasa) to 59.58% (Arusha)	30.29% (Mombasa) to 42.61% (Jinja)
Customer segment (3 segments)	57.14% (retail) to 59.95% (agent)	36.83% (retail) to 39.73% (merchant)

8 Fairness across subgroups

Approval rates and false-negative rates form a real but moderate spread rather than a stark disparity, though Mombasa and Nairobi stand out with approval rates roughly 7 points below the rest of the region set (52.15% and 52.33% respectively, against a cluster between 58–60% elsewhere), while Mombasa’s false-negative rate is simultaneously the lowest of the ten. Retail customers, who make up the large majority of test-set volume (25,081 of 31,212 transactions), are approved somewhat less often than agent or merchant customers (57.14% versus 59.95% and 59.23%) but are not the segment with the worst missed-fraud rate – merchant is (39.73% versus retail’s 36.83%).

The CBK Digital Credit Providers Regulations (2022) require lenders to give a clear, specific, substantiable reason when a decision goes against a customer. The SHAP-based local explanations

above are built for exactly this: each flagged transaction can be traced to its top contributing features in plain terms rather than treated as a black-box verdict. Before deployment, the Mombasa/Nairobi approval-rate gap and the merchant-segment false-negative gap would both warrant monitoring against regulatory scrutiny of disproportionate treatment, even where, as here, the gap looks driven by feature-distribution differences across regions rather than by a proxy for a protected attribute.

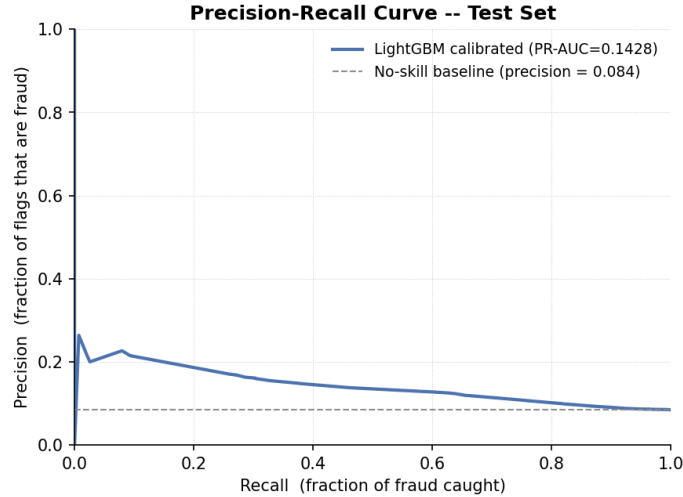


Figure 7: Precision-recall curve, tuned LightGBM, isotonic-calibrated, test set.

9 Recommendation

Seven sections ago, this exercise began with a model that looked, on the least demanding reading, almost adequate: over 91% accuracy on a dataset that is 91.6% not-fraud. Nothing in that number would have told the Head of Credit Risk what the institution actually needed to know – what a model costs when it is wrong, and how often it is wrong in the direction that matters. The work above exists to close that gap: to replace a plausible-sounding accuracy figure with a defensible, auditable, cost-based decision.

Three findings carried that work. First, the choice of ensemble family mattered less than expected – logistic regression, random forest, and LightGBM converged to a narrow PR-AUC band (0.126–0.132), which says more about the ceiling set by the available features than about any one algorithm. Second, imbalance handling done incorrectly does not merely inflate a metric modestly; resampling before the split more than doubled the reported PR-AUC on data the model had already partially seen, a result that would have looked like success and would have been a governance failure. Third, and most consequentially, an uncalibrated model is not simply a slightly worse model – it is a model whose scores cannot safely be read as probabilities, and a naive 0.5 cutoff applied to it, raw or calibrated, is either an accident that happens to work (Section 5) or an active liability that costs the institution KES 7.30 million more than it should (Section 6).

On that basis, this report recommends deployment of the Optuna-tuned LightGBM model (113 trees, 16 leaves, learning rate 0.0474), with isotonic calibration, at a decision threshold of $t \approx 0.080$, reviewed quarterly and re-derived whenever the fraud base rate or cost assumptions change materially. The uncalibrated model should not be used for threshold-based decisions under any circumstance: its scores are not on a probability scale, and a 0.5-style cutoff on them, however familiar, is arbitrary rather than principled. The calibrated model’s PR-AUC of 0.1428 against an 8.40% base rate shows real, if modest, discriminative power – sufficient to materially cut cost relative to a flat cutoff, but not sufficient on its own to replace manual review entirely for borderline cases, particularly in the Mombasa/Nairobi and merchant subgroups flagged in Section 7. The next material gain is more likely to come from richer behavioural features – rolling windows per `msisdn`,

which the current transaction-level snapshot cannot capture – than from a different algorithm.